



**COMPETENCY STANDARD
FOR
PRODUCTION PROCESS IN YARN MANUFACTURING
(TEXTILE SECTOR)**

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance
Government of the People's Republic of Bangladesh

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The Competency Standard for Production Process in Yarn Manufacturing is designed to guide the development of curricula, teaching materials, and assessment tools. It also serves as the document for providing training consistent with the requirements of the industry, so that individuals who pass through the set standard via assessment would be qualified and settled for a relevant job.

This document has been developed to be used in training under the Skills for Industry Competitiveness and Innovation Program (SICIP) to meet the skills requirements of the Textile sector. This document is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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Skills for Industry Competitiveness and Innovation Program (SICIP)

Finance Division, Ministry of Finance,

Probashi Kallyan Bhaban (Level-15 & 16)

71-72 Old Elephant Road, Eskaton Garden

Dhaka-1000

Phone: +8802 55138753-55, Fax: +88 02 55138752

Website: <https://sicip.gov.bd/>

INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades, which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The textile sector is the largest manufacturing and export-earning sector of Bangladesh, and yarn manufacturing (spinning) is the primary value-adding stage of the textile value chain. The productivity, quality and cost competitiveness of the entire downstream industry – weaving, knitting, wet processing and garment manufacturing – depend largely on the skills of the workforce engaged in the yarn production process. This Competency Standard has therefore been developed to prepare a competent entry-level workforce for the production process in yarn manufacturing.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and to ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competency and its corresponding Elements.

OVERVIEW:

A Competency Standard is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

1. provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
2. enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies.
3. encourages the development and delivery of flexible training that suits individual and industry requirements.
4. encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standards are developed by a working group comprised of national and international process experts and the participation of experts from the industry to identify the competencies required of an occupation in a particular sector.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at a training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

1. Reference to Industry Sector, Occupational Title and Occupational Description
2. Unit code
3. Unit title
4. Unit descriptor
5. Unit of Competency
6. Elements and performance criteria
7. Variables and range statement
8. Evidence guides

Together all the parts of a Unit of Competency:

1. Describe a work activity
2. Guide the assessor in determining whether the candidate is competent.

This Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competency:

1. An overview of all Units of Competency for the occupation and their corresponding duration required for completion of training.
2. The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

DESCRIPTION OF THE OCCUPATION AND THE TRAINING COURSE:

Industry Sector	Textile
Occupation Title	Production Process in Yarn Manufacturing
Occupational Description	A worker competent in the Production Process in Yarn Manufacturing carries out the sequential operations required to convert textile fibres into yarn in a spinning mill. The occupation covers mixing and blow room operations, carding, drawing, combing preparation and combing, roving (simplex), ring spinning, winding and yarn packing, together with in-process quality checking, routine machine care and housekeeping. The worker performs these tasks under supervision, following standard operating procedures, machine setting sheets, safety requirements and environmental practices of the mill, and records production and quality data using workplace formats.
Level of the Course	Basic Level Skills Training
Nominal Duration	360 Hours (03 Months)
Entry Qualification of Trainees	Grade VIII passed; physically and mentally fit to work in a spinning mill environment; minimum age as per the prevailing labour law of Bangladesh.
Target Number of Trainees	A total of 2,000 trainees to be trained by the end of 2028
Training Providers	Textile industries and training institutes having the capacity to deliver and run the training program in accordance with this Competency Standard and the Workshop/Lab Facility Standard.
Mode of Delivery	Institution-based and/or industry-based competency-based training combining theory, demonstration, guided practice and practical performance in an actual or simulated workplace.
Employment / Job Placement	The training providers (industries and training institutes) will arrange employment for at least 65% of the total enrollees upon completion of the training.
Assessment and Certification	Competency assessment will be conducted as per the Assessment Evidence Guide of each unit, in an assessment/training centre or in an actual or simulated workplace, by a nationally certified assessor or occupation-specific industry expert.
Possible Job Titles	Blow Room Operator/Helper, Card Tenter, Draw Frame Operator, Comber Operator, Simplex (Speed Frame) Operator, Ring Frame Sider/Doffer, Winding Operator, Yarn Packing Assistant, Quality Checker (Trainee level).

UNITS & ELEMENTS AT A GLANCE:

Generic Competencies (20 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-TEX-PPYM-01-G	Apply Occupational Health and Safety (OHS) Practices in the Workplace	1. Identify OHS policies and procedures 2. Apply personal health and safety practices 3. Report hazards and risks 4. Respond to emergencies	10
SICIP-TEX-PPYM-02-G	Work in a Team Environment	1. Identify team goals and processes 2. Communicate and cooperate with team members 3. Work as a team member 4. Solve problems as a team member	10
Total Hours			20 Hrs.

Sector Specific Competencies (20 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-TEX-PPYM-01-S	Work in the Textile Sector	1. Identify the textile sector and its main activities 2. Understand basic roles in textile production processes 3. Recognize common tools and equipment used 4. Understand simple quality requirements in textile work	10
SICIP-TEX-PPYM-02-S	Apply Green Practices in the Textile Sector	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
Total Hours			20 Hrs.

Occupation Specific Competencies (320 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-TEX-PPYM-01-O	Interpret Textile Fibres, Materials and Yarn Manufacturing Processes	1. Identify textile fibres and raw materials 2. Identify yarn types, counts and end uses 3. Identify machines, tools and equipment used in yarn manufacturing 4. Describe the yarn manufacturing process sequence 5. Interpret work instructions and production documents	30
SICIP-TEX-PPYM-02-O	Perform Mixing and Blow Room Operations	1. Prepare for blow room operations 2. Perform bale management, mixing and blending 3. Carry out opening and cleaning operations 4. Monitor lap/chute feed quality and machine performance	40
SICIP-TEX-PPYM-03-O	Perform Carding Operations	1. Prepare the carding machine for operation 2. Carry out the carding operation 3. Monitor card sliver quality and process parameters 4. Perform can changing, waste handling and machine cleaning	40
SICIP-TEX-PPYM-04-O	Perform Drawing and Combing Operations	1. Prepare the draw frame for operation 2. Carry out breaker and finisher drawing operations 3. Perform lap preparation for combing 4. Carry out combing operations 5. Check sliver quality and record process data 6. Develop concept of Auto-leveler 7. Handle machine and product faults, wastages and cleaning	50

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-TEX-PPYM-05-O	Perform Roving (Simplex) Operations	1. Prepare the speed frame for operation 2. Carry out the roving operation 3. Perform creeling, piecing and doffing 4. Check roving quality and maintain records	40
SICIP-TEX-PPYM-06-O	Perform Ring Spinning Operations	1. Prepare the ring frame for operation 2. Carry out the ring spinning operation 3. Perform piecing, creeling and doffing 4. Monitor yarn quality and control end breakages 5. Handle waste, cleaning and shift records	60
SICIP-TEX-PPYM-07-O	Perform Winding and Yarn Packing Operations	1. Prepare the winding machine for operation 2. Carry out winding operations 3. Perform splicing/knotting and yarn clearing 4. Carry out conditioning, packing and dispatch preparation	40
SICIP-TEX-PPYM-08-O	Perform In-process Quality Checks and Routine Machine Care	1. Perform in-process quality checks and basic tests 2. Record and report production and quality data 3. Carry out cleaning and routine preventive maintenance 4. Maintain housekeeping and workplace conditions	20
Total Hours			320 Hrs.

Grand Total Duration of the Training Course	360 Hours (03 Months)
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COMPETENCY STANDARD: PRODUCTION PROCESS IN YARN MANUFACTURING

The Generic Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES IN THE WORKPLACE	Nominal Duration: 10 Hrs.	Unit Code: SICIP-TEX-PPYM-01-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply Occupational Health and Safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures; applying personal health and safety practices; reporting hazards and risks; and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1. 1 <u>OHS policies</u> and safe operating procedures are read and understood. 1. 2 Safety signs and symbols are identified and followed. 1. 3 Emergency response, evacuation procedures and other contingency measures are determined.
2. Apply personal health and safety practices	2. 1 OHS policies and procedures are followed and practiced. 2. 2 <u>Personal Protective Equipment (PPE)</u> is selected and used. 2. 3 Personal hygiene is maintained.
3. Report hazards and risks	3.1 <u>Hazards and risks</u> are identified, assessed and controlled. 3.2 Incidents arising from hazards and risks are reported to the authority. 3.3 Corrective actions are implemented to correct unsafe conditions in the workplace.
4. Respond to emergencies	4. 1 Alarms and warning devices are responded to. 4. 2 <u>Emergency response plans and procedures</u> are implemented. 4. 3 <u>First aid procedure</u> is applied during emergency situations.

Range of Variables:

Variable	Range (May include but not limited to)
1. OHS policies	1. 1 International OHS requirements

Variable	Range (May include but not limited to)
	1. 2 Bangladesh standards for OHS 1. 3 Bangladesh Labour Act and Rules 1. 4 Building Code 1. 5 Fire Safety Rules and Regulations 1. 6 Industry guidelines and mill safety manual
2. Personal Protective Equipment (PPE)	2. 1 Nose/face mask (dust mask) 2. 2 Ear plugs or ear muffs 2. 3 Hand gloves 2. 4 Safety shoes 2. 5 Head cap/helmet 2. 6 Apron and overalls 2. 7 Goggles and safety glasses 2. 8 Hair net
3. Hazards and risks	3.1 Physical hazards (moving machine parts, nip points, noise, vibration, heat) 3.2 Chemical hazards (lubricants, cleaning agents, solvents) 3.3 Biological hazards (bacteria, fungus in damp areas) 3.4 Ergonomic hazards (repetitive work, standing, manual lifting) 3.5 Electrical hazards 3.6 Fire hazards (cotton dust, fly accumulation, static electricity)
4. Emergency response plans and procedures	4.1 Firefighting procedures 4.2 Earthquake response procedures 4.3 Evacuation procedures 4.4 Medical and first aid 4.5 Electrical accident response 4.6 Emergency machine stop procedures
5. First aid procedure	5. 1 Washing of open wound 5. 2 Washing chemically infected area 5. 3 Applying bandage 5. 4 Tourniquet 5. 5 Applying CPR (Cardiopulmonary Resuscitation) 5. 6 Taking appropriate medicine

Curricular Content Guide:

1. Underpinning Knowledge	1.1 OHS workplace policies and procedures. 1.2 Work safety procedures in a textile mill. 1.3 Emergency procedures. 1.3.1 Firefighting 1.3.2 Earthquake response. 1.3.3 Explosion response. 1.3.4 Accident response.
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	1.4 Types of hazards (biological, chemical, physical, ergonomic and electrical) and their effects. 1.5 PPE types and uses. 1.6 Personal hygiene practices. 1.7 Safety signs, symbols and colour codes. 1.8 OHS awareness and reporting requirements.
2. Underpinning Skills	2. 1 Identifying OHS policies and procedures 2. 2 Following personal work safety practices 2. 3 Selecting and using appropriate PPE 2. 4 Reporting hazards and risks 2. 5 Responding to emergency procedures 2. 6 Maintaining physical well-being in the workplace 2. 7 Performing basic firefighting using fire extinguishers 2. 8 Applying basic first aid procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 PPEs 4.3 Firefighting equipment 4.4 Emergency response manual 4.5 Safety signs and symbols 4.6 First aid kits

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Followed OHS policies and procedures. 1.2 Selected and used personal protective equipment (PPE). 1.3 Reported incidents arising from hazards and risks to the authority. 1.4 Implemented emergency response plans and procedures. 1.5 Applied basic first aid procedure.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning

	2. 4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: WORK IN A TEAM ENVIRONMENT	Nominal Duration: 10 Hrs.	Unit Code: SICIP-TEX-PPYM-02-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to work in a team environment. It specifically includes the tasks of identifying team goals and processes, communicating and cooperating with team members, working as a team member and solving problems as a team member.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and processes	1.1 Team goals and collaborative decision-making processes are identified. 1.2 Roles and responsibilities of team members are identified. 1.3 Relationships within the team and with other workers are identified.
2. Communicate and cooperate with team members	2. 1 Effective interpersonal skills are used to interact with team members and to contribute to activities and objectives. 2. 2 Formal and informal <u>forms of communication</u> are used effectively to support team achievement. 2. 3 Diversity in character is respected and valued in team functioning. 2. 4 Views and opinions of other team members are understood and valued. 2. 5 Workplace terminology is used correctly to assist communication.
3. Work as a team member	3.1 Duties, responsibilities, authorities, objectives and task requirements are identified and clarified with the team. 3.2 Tasks are performed in accordance with organizational and team requirements, specifications, and workplace procedures. 3.3 Support to other team members is provided to ensure the team achieves its goals. 3.4 Agreed reporting lines are followed using standard operating procedures.
4. Solve problems as a team member	4.1 Current and potential problems faced by the team are identified. 4.2 A solution to the problem is identified. 4.3 Problems are solved effectively and the outcome of the implemented solution is evaluated.

Range of Variables:

Variable	Range (May include but not limited to)
1. Forms of communication	1. 1 Agenda 1. 2 Simple reports such as progress and incident reports 1. 3 Job sheets and shift handover log 1. 4 Operational manuals and SOPs 1. 5 Brochures and promotional material 1. 6 Visual and graphic materials 1. 7 Standards 1. 8 OSH information 1. 9 Signs and symbols

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Team goals and collaborative decision-making processes 1. 2 Roles and responsibilities of team members 1. 3 Relationships within the team and with other workers 1. 4 Effective interpersonal skills to interact with team members 1. 5 Effective formal and informal forms of communication 1. 6 Value of diversity in team functioning 1. 7 Correct use of workplace terminology 1. 8 Team's duties, responsibilities, authorities, objectives and task requirements 1. 9 Support mechanisms for other members of the team to ensure achievement of goals 1. 10 Methods of identifying current and potential problems faced by a team 1. 11 Effective problem-solving methods and evaluation of outcomes
2. Underpinning Skills	2. 1 Identifying team goals and collaborative decision-making processes 2. 2 Identifying roles and responsibilities of team members 2. 3 Using effective interpersonal skills to interact with team members 2. 4 Using formal and informal forms of communication 2. 5 Understanding and valuing views and opinions of other team members 2. 6 Performing tasks in accordance with organizational and team requirements, specifications and workplace procedures 2. 7 Supporting other members of the team to ensure the team achieves its goals 2. 8 Identifying current and potential problems faced by the team 2. 9 Identifying solutions to problems

	2. 10 Solving problems effectively and evaluating the outcome of the implemented solution
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Manuals/Guidelines 4.3 Pens 4.4 Papers 4.5 Workbooks

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 Identified team goals and work processes. 1. 2 Communicated and cooperated with team members. 1. 3 Worked as a team member. 1. 4 Solved problems as a team member.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 4. Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

Unit of Competency: WORK IN THE TEXTILE SECTOR	Nominal Duration: 10 Hrs.	Unit Code: SICIP-TEX-PPYM-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required for work in the textile sector. It specifically includes the tasks of identifying the textile sector and its main activities, understanding basic roles in textile production processes, recognizing common tools and equipment used, and understanding simple quality requirements in textile work.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify the textile sector and its main activities	1. 1 The textile sector is identified within the context of national and global industries. 1. 2 <u>Main activities</u> of the textile sector are recognized. 1. 3 <u>Different stages of the textile value chain</u> are described in simple terms. 1. 4 Key workplaces in the textile sector are identified. 1. 5 Importance of the textile sector in the economy and the environment is understood.
2. Understand basic roles in textile production processes	2. 1 <u>Basic job roles in textile production</u> are identified and described. 2. 2 Simple production workflows and task sequences are understood. 2. 3 Teamwork and coordination in production processes are identified. 2. 4 Workplace safety roles and responsibilities are understood and followed. 2. 5 Shift working practices and reporting lines are understood.
3. Recognize common tools and equipment used	3.1 <u>Common tools used in textile processing</u> are identified correctly. 3.2 <u>Basic machines and equipment used in production</u> are recognized. 3.3 Functions of tools and equipment are understood in simple terms. 3.4 Basic maintenance and cleaning needs of tools and equipment are recognized.
4. Understand simple quality requirements in textile work	4.1 Basic quality standards in textile work are identified. 4.2 <u>Simple defects in textile products</u> are recognized. 4.3 Importance of maintaining quality during production is understood.

Elements of Competency	Performance Criteria
	4.4 Work output is checked against basic quality expectations. 4.5 Good workmanship practices are followed to ensure quality.

Range of Variables:

Variable	Range (May include but not limited to)
1. Main activities	1. 1 Fibre production and preparation 1. 2 Yarn manufacturing (spinning) 1. 3 Fabric manufacturing (weaving and knitting) 1. 4 Wet processing (dyeing, printing and finishing) 1. 5 Garment manufacturing 1. 6 Testing and quality assurance 1. 7 Packaging, storage and export
2. Different stages of the textile value chain	2. 1 Natural and man-made fibre production 2. 2 Fibre preparation and mixing 2. 3 Yarn manufacturing (spinning) 2. 4 Fabric manufacturing (weaving/knitting) 2. 5 Dyeing, printing and finishing 2. 6 Cutting, sewing and finishing of garments 2. 7 Packaging, distribution and export
3. Basic job roles in textile production	3.1 Blow room operator/helper 3.2 Card tenter 3.3 Draw frame and comber operator 3.4 Simplex (speed frame) operator 3.5 Ring frame sider and doffer 3.6 Winding and packing operator 3.7 Quality checker/laboratory assistant 3.8 Maintenance fitter/electrician 3.9 Line leader/supervisor
4. Common tools used in textile processing	4.1 Sliver and roving cutter, scissors and nippers 4.2 Hooks, brushes and dusting tools 4.3 Spanner, screwdriver and plier set 4.4 Oil can and grease gun 4.5 Feeler and setting gauges 4.6 Wrap reel, wrap block and electronic balance 4.7 Tachometer and hand lens 4.8 Vacuum cleaner and air blowing gun
5. Basic machines and equipment used in production	5. 1 Bale opener/plucker and blow room line 5. 2 Carding machine 5. 3 Draw frame (breaker and finisher) 5. 4 Lap former and comber 5. 5 Speed frame (simplex)

Variable	Range (May include but not limited to)
	5. 6 Ring frame 5. 7 Winding machine/autoconer 5. 8 Humidification and waste collection plant 5. 9 Testing laboratory equipment
6. Simple defects in textile products	6. 1 Neps and slubs 6. 2 Thick and thin places 6. 3 Count variation and uneven yarn 6. 4 High hairiness 6. 5 Foreign fibre and contamination 6. 6 Sliver and roving breakage 6. 7 Soft/hard and damaged packages 6. 8 Oil and dirt stains

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Structure and importance of the textile sector in the national and global economy. 1. 2 Main activities and stages of the textile value chain. 1. 3 Position of yarn manufacturing within the textile value chain. 1. 4 Basic job roles, responsibilities and reporting lines in textile production. 1. 5 Simple production workflows, task sequences and shift practices. 1. 6 Common tools, machines and equipment used in textile production and their functions. 1. 7 Basic maintenance and cleaning needs of tools and equipment. 1. 8 Basic quality standards and common defects in textile products. 1. 9 Importance of quality, productivity and good workmanship in the workplace.
2. Underpinning Skills	2. 1 Identifying the textile sector and its main activities 2. 2 Describing the stages of the textile value chain in simple terms 2. 3 Understanding basic job roles and responsibilities in textile production processes 2. 4 Following simple production workflows, task sequences and teamwork practices 2. 5 Applying workplace safety roles and responsibilities in production activities 2. 6 Recognizing common tools, machines and equipment and their functions 2. 7 Recognizing basic maintenance and cleaning needs of tools and equipment

	2. 8 Recognizing simple defects and checking work output against basic quality requirements 2. 9 Following good workmanship practices to ensure consistent quality output
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Charts, models and samples of fibre, sliver, roving and yarn 4.3 Multimedia projector, laptop and audio-visual clips 4.4 Learning manual and handouts 4.5 Stationery

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 Identified the textile sector and its main activities. 1. 2 Understood basic roles in textile production processes. 1. 3 Recognized common tools and equipment used. 1. 4 Understood simple quality requirements in textile work.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY GREEN PRACTICES IN THE TEXTILE SECTOR	Nominal Duration: 10 Hrs.	Unit Code: SICIP-TEX-PPYM-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply green practices in the textile sector. It specifically includes the tasks of interpreting green concepts, minimizing resource use in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1. 1 Basic concepts of environmental sustainability and green practices are explained clearly. 1. 2 <u>Environmental impacts of textile production</u> processes are identified. 1. 3 <u>Resource efficiency concepts</u> are interpreted. 1. 4 Waste reduction, reuse and recycling principles are described in relation to yarn manufacturing operations. 1. 5 <u>Eco-friendly practices in the textile sector</u> are recognized and related to workplace activities.
2. Minimize resource use in the workplace	2. 1 Electricity, water, compressed air and raw materials are used efficiently according to workplace requirements. 2. 2 <u>Unnecessary consumption of resources</u> is identified and avoided during operations. 2. 3 Machines and equipment are operated in energy-efficient modes where applicable. 2. 4 Material wastage is minimized through careful handling and accurate measurement. 2. 5 <u>Resource-saving practices</u> are applied where possible. 2. 6 Leakage, spillage and wastage of materials are promptly controlled and reported. 2. 7 <u>Workplace practices</u> for resource conservation are followed.
3. Implement waste management practices	3.1 <u>Types of waste</u> generated in yarn manufacturing operations are identified correctly. 3.2 Waste materials are segregated according to workplace and environmental requirements. 3.3 <u>Reusable and recyclable materials</u> are separated from non-recyclable waste. 3.4 Waste is stored safely in designated containers or areas. 3.5 Waste disposal is carried out following environmental rules and organizational procedures.

Elements of Competency	Performance Criteria
	3.6 Hazardous waste is handled carefully according to safety guidelines. 3.7 7. Waste reduction practices are applied to minimize environmental impact during operations.

Range of Variables:

Variable	Range (May include but not limited to)
1. Environmental impacts of textile production	1. 1 High electrical energy consumption of spinning machinery and humidification plants 1. 2 Cotton dust, fibre fly and micro-fibre emission 1. 3 Noise pollution from machines and compressors 1. 4 Solid waste generation (droppings, comber noil, soft and hard waste) 1. 5 Water consumption and effluent from wet processing 1. 6 Air emission and greenhouse gases from boilers and generators 1. 7 Used oil, grease and contaminated cotton waste
2. Resource efficiency concepts	2. 1 Energy-efficient motors and inverter/variable frequency drives 2. 2 Control of compressed air leakage and optimized air pressure 2. 3 Optimized humidification and ventilation control 2. 4 Reduction of soft and hard waste percentage 2. 5 Preventive maintenance to reduce power consumption 2. 6 Energy-efficient lighting and switching-off practices 2. 7 Optimum machine loading and reduction of idle running
3. Eco-friendly practices in the textile sector	3.1 Recycling of soft waste back into mixing where permitted 3.2 Use of recycled, organic or sustainably sourced fibre 3.3 Dust collection, filter and waste recovery systems 3.4 Compliance with Department of Environment (DoE) rules and regulations 3.5 Effluent treatment plant (ETP) and rainwater harvesting 3.6 Reuse of cans, bobbins, cones, cartons and packaging materials 3.7 Green building and energy audit practices
4. Unnecessary consumption of resources	4.1 Idle running of machines and motors 4.2 Compressed air leakage in pipelines and guns 4.3 Excess lighting in unoccupied areas 4.4 Over-cleaning with air guns instead of vacuum 4.5 Excess lubricant and cleaning material use 4.6 Excessive good fibre loss with droppings and noil

Variable	Range (May include but not limited to)
5. Resource-saving practices	5. 1 Switching off machines, lights and fans when not required 5. 2 Reporting air, water and oil leakage promptly 5. 3 Using vacuum cleaning instead of air blowing where possible 5. 4 Reusing cans, tubes, cones and packaging materials 5. 5 Maintaining correct machine settings to reduce waste 5. 6 Recording and monitoring waste and energy use
6. Workplace practices	6. 1 Standard operating procedures (SOPs) 6. 2 Environmental policy of the organization 6. 3 Housekeeping and 5S practices 6. 4 Energy and waste monitoring records 6. 5 Instructions from supervisors and safety notices
7. Types of waste	7.1 Blow room droppings and card droppings/flat strips 7.2 Comber noil 7.3 Sliver, roving and pneumafil soft waste 7.4 Hard waste from winding (yarn and thread waste) 7.5 Damaged cans, bobbins, cones and tubes 7.6 Packaging waste (poly bags, bale wrappers, straps, cartons) 7.7 Used oil, grease and oil-contaminated cotton (hazardous waste) 7.8 Used filters, sweeps and floor waste
8. Reusable and recyclable materials	8.1 Usable soft waste for back-mixing 8.2 Comber noil for coarse count and open-end spinning 8.3 Cans, bobbins, tubes and cones 8.4 Cartons, poly bags and bale straps 8.5 Metal and plastic scrap 8.6 Paper and stationery waste

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Basic concepts of environmental sustainability, green practices and climate-resilient production. 1. 2 Environmental impacts of textile and yarn manufacturing operations. 1. 3 Resource efficiency concepts relating to energy, water, compressed air and raw material. 1. 4 Principles of waste reduction, reuse and recycling in a spinning mill. 1. 5 Eco-friendly practices and technologies applied in the textile sector. 1. 6 Types of waste generated at each stage of yarn manufacturing and permissible waste levels.
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	1. 7 Waste segregation, storage, disposal methods and compliance with environmental regulations. 1. 8 Hazardous waste handling and workplace environmental safety procedures. 1. 9 Relevant national environmental rules and organizational environmental policy.
2. Underpinning Skills	2. 1 Interpreting green concepts and relating them to workplace activities 2. 2 Identifying environmental impacts of yarn manufacturing operations 2. 3 Applying resource efficiency principles in workplace operations 2. 4 Using electricity, compressed air, water and raw materials efficiently 2. 5 Identifying and avoiding unnecessary consumption of resources 2. 6 Operating machines and equipment in energy-efficient modes where applicable 2. 7 Handling materials carefully to reduce wastage and improve accuracy 2. 8 Controlling and reporting leakage, spillage and resource wastage promptly 2. 9 Identifying and segregating different types of waste generated in yarn manufacturing 2. 10 Separating reusable and recyclable materials from non-recyclable waste 2. 11 Handling hazardous waste according to safety guidelines and regulations
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4. 1 Workplace (simulated or actual) 4. 2 Standard operating procedures 4. 3 Workplace documents, signs and symbols 4. 4 Codes of conduct and environmental policy 4. 5 Waste segregation bins and storage area 4. 6 Tools, equipment and facilities appropriate to the process or activities 4. 7 Materials relevant to the proposed activity

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 Interpreted green concepts. 1. 2 Minimized resource use in the workplace. 1. 3 Implemented waste management practices.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: INTERPRET TEXTILE FIBRES, MATERIALS AND YARN MANUFACTURING PROCESSES	Nominal Duration: 30 Hrs.	Unit Code: SICIP-TEX-PPYM-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to interpret textile fibres, materials and yarn manufacturing processes. It specifically includes the tasks of identifying textile fibres and raw materials, identifying yarn types, counts and end uses, identifying machines, tools and equipment used in yarn manufacturing, describing the yarn manufacturing process sequence, and interpreting work instructions and production documents.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify textile fibres and raw materials	1. 1 <u>Types of textile fibres</u> are identified based on source and characteristics. 1. 2 Physical properties of textile fibres are described. 1. 3 <u>Fibre quality parameters</u> are recognized and related to yarn quality. 1. 4 <u>Raw material documents</u> and bale/lot identification marks are interpreted. 1. 5 <u>Contamination and foreign matter</u> in raw material are recognized.
2. Identify yarn types, counts and end uses	2. 1 <u>Types of yarn</u> produced in a spinning mill are identified. 2. 2 <u>Yarn count systems</u> are interpreted and simple conversions are performed. 2. 3 <u>Yarn quality characteristics</u> are described. 2. 4 Twist and twist multiplier are explained in relation to yarn type. 2. 5 End uses of different yarns are identified.
3. Identify machines, tools and equipment used in yarn manufacturing	3.1 <u>Machines used in yarn manufacturing</u> are identified. 3.2 <u>Hand tools and measuring instruments</u> are identified and their functions are described. 3.3 Machines are classified according to process stages. 3.4 Basic specifications and operational features of machines are recognized. 3.5 Safety requirements for using machines, tools and equipment are identified.
4. Describe the yarn manufacturing process sequence	4. 1 <u>Stages of the yarn manufacturing process</u> are described in the correct sequence. 4. 2 Objectives of each process stage are explained.

Elements of Competency	Performance Criteria
	4. 3 <u>Material flow and in-process material forms</u> are identified. 4. 4 <u>Process parameters</u> influencing yarn quality are described. 4. 5 Differences between carded, combed and blended yarn manufacturing routes are explained.
5. Interpret work instructions and production documents	5.1 <u>Production documents</u> are identified and interpreted. 5.2 Job/work orders, lot plans and machine setting sheets are read and understood. 5.3 Standard operating procedures (SOPs) are followed as instructed. 5.4 Information is recorded and reported using workplace formats. 5.5 Workplace terminology, units and symbols are used correctly.

Range of Variables:

Variable	Range (May include but not limited to)
1. Types of textile fibres	1.1 Natural fibres 1.1.1 Cotton 1.1.2 Jute 1.1.3 Wool 1.1.4 Silk 1.2 Man-made fibres 1.2.1 Polyester 1.2.2 Acrylic 1.2.3 Nylon 1.2.4 Polypropylene 1.3 Regenerated fibres 1.3.1 Viscose rayonModal 1.3.2 Lyocell/Tencel 1.4 Blends 1.4.1 Polyester-cotton (PC) 1.4.2 Chief value cotton (CVC) 1.4.3 Poly-viscose (PV) 1.4.4 Cotton-viscose 1.5 Recycled and organic fibres
2. Fibre quality parameters	2. 1 Staple length and length uniformity 2. 2 Fineness/micronaire value 2. 3 Fibre strength and elongation 2. 4 Maturity ratio 2. 5 Trash and dust content 2. 6 Short fibre content

Variable	Range (May include but not limited to)
	2. 7 Neps per gram 2. 8 Colour and grade 2. 9 Moisture regain
3. Raw material documents	3.1 Bale/lot card and bale identification tag 3.2 Mixing plan and laydown chart 3.3 HVI/fibre test report 3.4 Store requisition and gate pass 3.5 Supplier invoice/packing list
4. Contamination and foreign matter	4. 1 Polypropylene and plastic strings 4. 2 Jute and cloth pieces 4. 3 Human and animal hair, feathers 4. 4 Coloured and dyed fibres 4. 5 Metal particles and stones 4. 6 Oil-stained fibre 4. 7 Seed coat fragments and trash
5. Types of yarn	5. 1 Carded yarn 5. 2 Combed yarn 5. 3 Blended yarn 5. 4 Single and ply (doubled) yarn 5. 5 Ring spun yarn 5. 6 Rotor/open-end yarn 5. 7 Compact yarn 5. 8 Slub and fancy yarn
6. Yarn count systems	6.1 Indirect systems 6.1.1 English count (Ne) 6.1.2 Metric count (Nm) 6.1.3 Worsted count 6.2 Direct systems 6.2.1 Tex 6.2.2 Denier 6.2.3 Decitex 6.3 Conversion relations between count systems 6.4 Twist per inch (TPI) and twist multiplier (TM)
7. Yarn quality characteristics	7.1 Count and count variation (CV%) 7.2 Unevenness (U%/CV%) 7.3 Imperfections (thin places, thick places, neps) 7.4 Single yarn strength, CSP and RKM 7.5 Elongation 7.6 Hairiness 7.7 Classimat faults 7.8 Appearance grade
8. Machines used in yarn manufacturing	8.1 Bale opener/plucker 8.2 Blow room line (mixer, cleaner, condenser, chute feed)

Variable	Range (May include but not limited to)
	8.3 Carding machine 8.4 Draw frame (breaker and finisher) 8.5 Lap former/unilap machine 8.6 Comber machine 8.7 Speed frame (simplex) 8.8 Ring frame 8.9 Winding machine/autoconer 8.10 Doubling and TFO machine (where applicable) 8.11 Humidification and waste collection plant 8.12 Yarn conditioning and packing/baling press
9. Hand tools and measuring instruments	9.1 Hooks, brushes and dusting tools 9.2 Spanner, screwdriver and plier set 9.3 Oil can and grease gun 9.4 Feeler and setting gauges 9.5 Wrap reel and wrap block 9.6 Electronic balance 9.7 Tachometer 9.8 Hand lens/magnifier 9.9 Hygrometer, thermometer and moisture meter 9.10 Vacuum cleaner and air blowing gun 9.11 Belt joining machine
10. Stages of the yarn manufacturing process	10.1 Mixing and blow room (opening, cleaning and blending) 10.2 Carding 10.3 Drawing – breaker passage 10.4 Combing preparation (lap forming) and combing 10.5 Drawing – finisher passage 10.6 Roving/simplex (speed frame) 10.7 Ring spinning 10.8 Winding 10.9 Yarn conditioning, and packing
11. Material flow and in-process material forms	11.1 Bale → tuft/flock 11.2 Lap or chute feed 11.3 Card sliver 11.4 Drawn sliver (breaker and finisher) 11.5 Lap for combing and comber sliver 11.6 Roving 11.7 Ring cop/bobbin 11.8 Cone/yarn package 11.9 Packed carton/bag
12. Process parameters	12. 1 Draft (break draft, main draft, total draft) 12. 2 Doubling 12. 3 Twist per inch (TPI) and twist multiplier 12. 4 Machine speed (rpm, m/min) 12. 5 Hank/count of the material

Variable	Range (May include but not limited to)
	12. 6 Production rate and machine efficiency 12. 7 Waste and noil percentage 12. 8 Relative humidity and temperature 12. 9 Machine settings and gauges
13. Production documents	13. 1 Shift and daily production report 13. 2 Lot plan and mixing plan 13. 3 Machine setting sheet 13. 4 Quality/test report 13. 5 Waste report 13. 6 Maintenance checklist 13. 7 Breakdown/downtime record 13. 8 Log book and shift handover register

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Fibre quality parameters and their influence on yarn quality and processing performance. 1.2 Contamination types, their sources and their effect on yarn and fabric quality. 1.3 Yarn types, count systems, count conversion, twist and twist multiplier. 1.4 Yarn quality characteristics and their relationship with end use. 1.5 Machines used at each stage of yarn manufacturing and their functions. 1.6 Sequence and objectives of the yarn manufacturing process. 1.7 In-process material forms and material flow through the spinning line. 1.8 Process parameters (draft, doubling, twist, speed) and their influence on quality. 1.9 Difference between carded, combed, blended and rotor-spun yarn routes. 1.10 Production documents, machine setting sheets, SOPs and workplace terminology.
2. Underpinning Skills	2. 1 Identifying textile fibres by appearance and simple physical tests 2. 2 Reading bale/lot cards, mixing plans and fibre test reports 2. 3 Recognizing contamination and foreign matter in raw material 2. 4 Identifying yarn types and determining yarn count using wrap reel and balance

	2. 5 Performing simple count conversions and twist calculations 2. 6 Identifying machines, tools and measuring instruments used in yarn manufacturing 2. 7 Describing the yarn manufacturing process sequence and objectives of each stage 2. 8 Identifying in-process material forms and material flow 2. 9 Interpreting work orders, machine setting sheets and production documents
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4. 1 Workplace (simulated or actual) 4. 2 Fibre sample board and fibre samples (natural, man-made and blends) 4. 3 Samples of lap, sliver, roving, ring cop and cone 4. 4 Yarn samples of different counts and types 4. 5 Charts, models and flow diagrams of the yarn manufacturing process 4. 6 Wrap reel, wrap block and electronic balance 4. 7 Hand tools and measuring instruments 4. 8 Multimedia projector, laptop and audio-visual clips 4. 9 Learning manual, SOPs and sample production formats

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Identified textile fibres and raw materials. 1.2 Identified yarn types, counts and end uses. 1.3 Identified machines, tools and equipment used in yarn manufacturing. 1.4 Described the yarn manufacturing process sequence. 1.5 Interpreted work instructions and production documents.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)

3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.
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Unit of Competency: PERFORM MIXING AND BLOW ROOM OPERATIONS	Nominal Duration: 40 Hrs.	Unit Code: SICIP-TEX-PPYM-02-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform mixing and blow room operations. It specifically includes the tasks of preparing for blow room operations, performing bale management, mixing and blending, carrying out opening and cleaning operations, and monitoring lap/chute feed quality and machine performance.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for blow room operations	1.1 Personal protective equipment (PPE) is selected and used in accordance with safety guidelines. 1.2 Work order, <u>mixing plan</u> and lot details are obtained and understood. 1.3 Machines and work area are checked for cleanliness and readiness. 1.4 <u>Safety devices and guards</u> are checked before start-up. 1.5 Required <u>raw materials and accessories</u> are arranged at the workstation. 1.6 Abnormal conditions found during checking are reported to the authority.
2. Perform bale management, mixing and blending	2. 1 Bales are identified, positioned and opened following the <u>laydown plan</u> . 2. 2 Bale wrappers, straps and <u>contaminations</u> are removed and disposed of properly. 2. 3 Fibre layers are taken uniformly from all bales to maintain homogeneous mixing. 2. 4 <u>Mixing and blending methods</u> are applied as per lot requirement. 2. 5 Blend proportion is maintained according to specification. 2. 6 Conditioning of fibre is ensured before feeding to the machine. 2. 7 Soft waste is added to the mixing within the permissible limit as instructed.
3. Carry out opening and cleaning operations	3.1 Blow room machines are started and stopped following <u>standard operating procedures</u> . 3.2 <u>Machine settings</u> are adjusted according to material and process requirements. 3.3 Feed rate and material flow are maintained uniformly through the line.

Elements of Competency	Performance Criteria
	3.4 <u>Beater and cleaning points</u> are monitored during operation. 3.5 Choking, lapping and material accumulation are cleared safely. 3.6 Droppings and waste are collected and disposed of as per workplace procedure. 3.7 Fire and spark protection devices are kept functional during operation.
4. Monitor lap/chute feed quality and machine performance	4. 1 <u>Lap/chute feed quality parameters</u> are checked at specified intervals. 4. 2 Lap weight, lap variation and chute feed uniformity are measured and recorded. 4. 3 <u>Common faults</u> are identified and reported to the authority. 4. 4 Waste percentage is checked against the standard. 4. 5 Production and quality data are recorded in the <u>prescribed format</u>.
5. Perform machine cleaning	5. 1 Machine cleaning requirements are identified according to machine type and workplace procedures. 5. 2 Machine is stopped, isolated and made safe before cleaning according to safety procedures. 5. 3 Dust, lint, fibre waste and other contaminants are removed using appropriate cleaning tools and methods. 5. 4 Machine condition is checked after cleaning and any abnormality is reported according to workplace procedures.

Range of Variables:

Variable	Range (May include but not limited to)
1. Mixing plan	1.1 Lot number and mixing number 1.2 Fibre type, origin and grade 1.3 Blend ratio and blend components 1.4 Number of bales in the laydown 1.5 Percentage of soft waste to be added 1.6 Target yarn count and quality 1.7 Total mixing weight
2. Safety devices and guards	2. 1 Emergency stop buttons and pull cords 2. 2 Machine guards, covers and interlock switches 2. 3 Metal, spark and fire detection/diverter system 2. 4 Earthing and electrical safety devices 2. 5 Warning signs and floor markings 2. 6 Lockout–tagout arrangement

Variable	Range (May include but not limited to)
3. Raw materials and accessories	3.1 Cotton and man-made fibre bales 3.2 Soft waste for re-mixing 3.3 Lap rods and lap spools 3.4 Sliver cans and trolleys 3.5 Bale cutter and strap cutter 3.6 Cleaning tools, brushes and vacuum cleaner 3.7 Bobbin 3.8 Paper cone
4. Laydown plan	4.1 Bale arrangement and stack pattern 4.2 Number of bales per mixing 4.3 Sequence of bale opening 4.4 Stack height and spacing 4.5 Position of blend components in the laydown
5. Contaminations	5.1 Polypropylene strings, plastic and poly bags 5.2 Jute and cloth pieces 5.3 Human and animal hair, feathers 5.4 Coloured and foreign fibres 5.5 Metal pieces, wire and stones 5.6 Oil-stained and mildewed fibre 5.7 Seed coat, leaf and trash particles
6. Mixing and blending methods	6.1 Hand/stack mixing 6.2 Bale plucker (automatic) mixing 6.3 Multi-mixer and blending machine 6.4 Chute feed blending 6.5 Back-mixing of soft waste 6.6 Weighing pan/hopper blending
7. Standard operating procedures	7.1 Start-up and shut-down sequence of the line 7.2 Safe feeding and material handling procedure 7.3 Cleaning schedule and method 7.4 Emergency stop procedure 7.5 Choking and lapping clearance procedure 7.6 Reporting and record keeping procedure
8. Machine settings	8.1 Beater speed and beater type 8.2 Feed roller setting and feed rate 8.3 Grid bar setting and grid angle 8.4 Fan speed and suction pressure 8.5 Opening roller/licker-in gauge 8.6 Cleaning intensity/cleaning point selection 8.7 Lap length and lap weight setting 8.8 Chute feed pressure setting
9. Beater and cleaning points	9.1 Bale opener beater and pin beater 9.2 Step cleaner and axial flow cleaner 9.3 Mono/multi cylinder cleaner

Variable	Range (May include but not limited to)
	9.4 Grid bars and mote knives 9.5 Dust cage and condenser 9.6 Filter unit and waste collection duct
10. Lap/chute feed quality parameters	10.1 Lap weight per unit length (g/m or g/yd) 10.2 Lap regularity/variation (A%) 10.3 Lap thickness 10.4 Chute feed pressure and uniformity across the card 10.5 Trash percentage after cleaning 10.6 Moisture content of the material 10.7 Cleaning efficiency of the line
11. Common faults	11.1 Conical, soft or hard lap 11.2 Uneven lap weight and lap variation 11.3 Licking and lapping of fibre on rollers/beaters 11.4 Choking of pipes and ducts 11.5 Excessive droppings and good fibre loss 11.6 Nep generation due to over-beating 11.7 Variation in chute feed across cards 11.8 Contamination carried into the line
12. Prescribed format	12. 1 Shift production report 12. 2 Lap weight/chute feed check sheet 12. 3 Waste record 12. 4 Machine stoppage/downtime record 12. 5 Cleaning checklist 12. 6 Contamination removal record

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Objectives and functions of mixing and the blow room process. 1. 2 Bale management, laydown planning and bale conditioning. 1. 3 Types of contamination, their sources and their effect on yarn quality. 1. 4 Sequence of blow room machines and the function of each machine. 1. 5 Machine settings and their effect on opening, cleaning and nep generation. 1. 6 Principle of "gentle opening and progressive cleaning". 1. 7 Fire and spark hazards in the blow room and their control. 1. 8 Lap and chute feed quality parameters and permissible tolerance. 1. 9 Humidity and conditioning requirements in the blow room. 1. 10 Cleaning schedules, housekeeping and record-keeping requirements.
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2. Underpinning Skills	2. 1 Interpreting mixing plans, laydown charts and work orders 2. 2 Opening bales and removing wrappers, straps and contamination 2. 3 Performing stack mixing and machine mixing to the specified blend ratio 2. 4 Adding soft waste to the mixing within permissible limits 2. 5 Starting and stopping blow room machines safely following SOPs 2. 6 Maintaining uniform feed rate and material flow 2. 7 Clearing choking, lapping and accumulation safely 2. 8 Measuring lap weight and lap variation and checking chute feed uniformity 2. 9 Identifying common blow room faults and reporting them 2. 10 Recording production and quality data in prescribed formats
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4. 1 Workplace (simulated or actual) 4. 2 Blow room line (bale opener, mixer, cleaner, condenser/chute feed) 4. 3 Cotton and man-made fibre bales and soft waste 4. 4 Bale cutter, strap cutter, trolleys, cans and lap rods 4. 5 Electronic balance and lap weighing scale 4. 6 Trash analyser 4. 7 Cleaning tools, brushes and industrial vacuum cleaner 4. 8 Learning manual and audio-visual clips

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 Prepared for blow room operations. 1. 2 Performed bale management, mixing and blending. 1. 3 Carried out opening and cleaning operations. 1. 4 Monitored lap/chute feed quality and machine performance.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration

	2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM CARDING OPERATIONS	Nominal Duration: 40 Hrs.	Unit Code: SICIP-TEX-PPYM-03-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform carding operations. It specifically includes the tasks of preparing the carding machine for operation, carrying out the carding operation, monitoring card sliver quality and process parameters, and performing can changing, waste handling and machine cleaning.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare the carding machine for operation	1. 1 Personal protective equipment (PPE) is selected and used in accordance with safety guidelines. 1. 2 Work order, lot details and the <u>machine setting sheet</u> are obtained and understood. 1. 3 <u>Card clothing and machine parts</u> are checked for condition and cleanliness. 1. 4 <u>Safety guards and devices</u> are checked before start-up. 1. 5 <u>Settings and gauges</u> are verified as per the requirement. 1. 6 Feed material (lap/chute feed) and empty cans are arranged at the machine.
2. Carry out the carding operation	2. 1 The machine is started and stopped following standard operating procedures. 2. 2 Feed material is fed uniformly to the machine. 2. 3 <u>Draft, speed and production parameters</u> are maintained as per requirement. 2. 4 Sliver is passed through the trumpet and coiler and coiled properly into the can. 2. 5 <u>Web and sliver appearance</u> is observed continuously during running. 2. 6 Sliver breakage and web fault are attended promptly and correctly. 2. 7 <u>Stoppages</u> are attended or reported according to the level of authority.
3. Monitor card sliver quality and process parameters	3.1 <u>Sliver quality parameters</u> are checked at specified intervals. 3.2 Sliver hank/weight is measured and recorded. 3.3 <u>Card sliver defects</u> are identified and reported. 3.4 Waste (droppings, flat strips and fly) is checked against the standard percentage. 3.5 Corrective action is taken or reported as per workplace procedure. 3.6 Data are recorded in the <u>prescribed format</u> .

Elements of Competency	Performance Criteria
4. Perform can changing, waste handling and machine cleaning	4. 1 Full cans are changed and replaced safely with correct <u>can identification</u>. 4. 2 Sliver ends are joined following the standard piecing method. 4. 3 Waste is collected, segregated, weighed and stored as per workplace procedure. 4. 4 <u>Cleaning of the machine and card clothing</u> is carried out following the schedule. 4. 5 Work area is kept clean and free from fly and fibre accumulation. 4. 6 Cleaning and maintenance activities are recorded in the checklist.

Range of Variables:

Variable	Range (May include but not limited to)
1. Machine setting sheet	1. 1 Hank of sliver to be delivered 1. 2 Total draft and draft distribution 1. 3 Cylinder, licker-in, flat and doffer speeds 1. 4 Gauge settings 1. 5 Delivery speed and production rate 1. 6 Permissible waste percentage
2. Card clothing and machine parts	2. 1 Cylinder and cylinder wire 2. 2 Licker-in (taker-in) and mote knives 2. 3 Flats and flat tops 2. 4 Doffer and doffer comb/stripping roller 2. 5 Feed roller and feed plate 2. 6 Crush rollers and web condenser 2. 7 Trumpet, calender rollers and coiler 2. 8 Can plate and can changer 2. 9 Suction/waste box and filter 2. 10 Auto-leveller (where fitted)
3. Safety guards and devices	3.1 Cylinder and doffer guards 3.2 Emergency stop button 3.3 Door interlocks and safety switches 3.4 Fire and spark protection device 3.5 Earthing and electrical protection 3.6 Warning signage
4. Settings and gauges	4. 1 Feed plate to licker-in setting 4. 2 Licker-in to cylinder setting 4. 3 Cylinder to flats setting 4. 4 Cylinder to doffer setting 4. 5 Mote knife setting 4. 6 Back plate and front plate setting

Variable	Range (May include but not limited to)
	4. 7 Trumpet size and coiler setting
5. Draft, speed and production parameters	5.1 Total draft 5.2 Cylinder speed (rpm) 5.3 Licker-in speed (rpm) 5.4 Flat speed (mm/min) 5.5 Doffer speed (rpm) 5.6 Delivery speed (m/min) 5.7 Hank/count of delivered sliver 5.8 Production per shift and machine efficiency
6. Web and sliver appearance	6.1 Uniformity and transparency of the web 6.2 Cloudy or patchy web 6.3 Holes and splits in the web 6.4 Neps and seed particles in the web 6.5 Sliver thickness, smoothness and coiling pattern
7. Stoppages	7.1 Sliver break and coiler choking 7.2 Can full stop and can change 7.3 Lapping on cylinder, doffer or licker-in 7.4 Web break and web sag 7.5 Auto-leveller alarm 7.6 Electrical trip and power failure 7.7 Safety door open
8. Sliver quality parameters	8.1 Hank/count of sliver (Ne or ktex) 8.2 Weight per unit length 8.3 Unevenness (U%/CV%) 8.4 Neps per gram 8.5 Trash and dust content 8.6 Sliver regularity and appearance 8.7 Cleaning efficiency of the card
9. Card sliver defects	9.1 Thick and thin places 9.2 Neps and seed coat fragments 9.3 Uneven and irregular sliver 9.4 Fluffy and hairy sliver 9.5 Cut, broken or stretched sliver 9.6 High trash and dust content 9.7 Contamination in the sliver
10. Can identification	10.1 Lot number and mixing number 10.2 Machine number and shift 10.3 Hank of sliver 10.4 Date and time of doffing 10.5 Colour coding of cans 10.6 Colour belt

Variable	Range (May include but not limited to)
11. Cleaning of the machine and card clothing	11.1 Stripping of cylinder, doffer and licker-in (brush/vacuum) 11.2 Cleaning of flats and flat strips removal 11.3 Grinding/buffing schedule (by authorized personnel) 11.4 Cleaning of under-casing and mote knife area 11.5 Cleaning of coiler, trumpet and calender rollers 11.6 Cleaning of suction ducts and filters 11.7 Floor and work area cleaning
12. Prescribed format	12.1 Sliver hank/weight check sheet 12.2 Shift production report 12.3 Waste record (droppings, flat strips, fly) 12.4 Stoppage/downtime record 12.5 Cleaning and maintenance checklist

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Construction and working principle of the carding machine. 1.2 Card clothing, wire points, their condition and maintenance requirements. 1.3 Function of licker-in, cylinder, flats, doffer, crush roller and coiler. 1.4 Settings and gauges of the card and their effect on quality and waste. 1.5 Auto-levelling principle and its benefit. 1.6 Web and sliver appearance and the interpretation of visual defects. 1.7 Card sliver quality parameters, neps, trash and cleaning efficiency. 1.8 Piecing method for sliver joining and can changing procedure. 1.9 Safety devices, safe operating and lockout–tagout procedures for the card. 1.10 Cleaning, stripping and grinding schedules and housekeeping requirements.
2. Underpinning Skills	2.1 Checking card clothing, machine parts and safety devices before start-up 2.2 Verifying settings and gauges under supervision 2.3 Starting and stopping the carding machine safely following SOPs 2.4 Maintaining uniform feed and monitoring web and sliver appearance 2.5 Piecing broken slivers and clearing lapping safely 2.6 Changing full cans and identifying cans correctly 2.7 Measuring sliver hank/weight and checking sliver quality 2.8 Identifying card sliver defects and their probable causes

	2. 9 Collecting, segregating, weighing and recording waste 2. 10 Recording production, quality and maintenance data
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Carding machine with accessories 4.3 Lap/chute feed material and sliver cans 4.4 Setting and feeler gauges 4.5 Electronic balance and sliver hank measuring arrangement 4.6 Trash analyser and nep counting board with hand lens 4.7 Tachometer 4.8 SOPs, machine setting sheets and recording formats 4.9 Learning manual and audio-visual clips

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Prepared the carding machine for operation. 1.2 Carried out the carding operation. 1.3 Monitored card sliver quality and process parameters. 1.4 Performed can changing, waste handling and machine cleaning.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM DRAWING AND COMBING OPERATIONS	Nominal Duration: 50 Hrs.	Unit Code: SICIP-TEX-PPYM-04-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform drawing and combing operations. It specifically includes the tasks of preparing the draw frame for operation, carrying out breaker and finisher drawing operations, performing lap preparation for combing, carrying out combing operations, and checking sliver quality and recording process data.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare the draw frame for operation	1.1 Personal protective equipment (PPE) is selected and used in accordance with safety guidelines. 1.2 Work order and machine setting sheet are obtained and understood. 1.3 <u>Drafting system parts</u> are checked for condition and cleanliness. 1.4 <u>Roller settings and drafts</u> are set/verified as per requirement. 1.5 The creel is loaded with the required number of <u>doublings</u> and cans are arranged properly. 1.6 Safety guards, <u>stop motions</u> and the <u>auto-leveller</u> are checked for proper functioning.
2. Carry out breaker and finisher drawing operations	2.1 The machine is started and stopped following standard operating procedures. 2.2 Slivers are passed through the <u>creel, drafting zone, trumpet and coiler</u> correctly. 2.3 <u>Process parameters</u> are maintained during the operation. 2.4 Breaker and finisher passages are completed as per the lot plan. 2.5 Sliver breakage and roller lapping are attended promptly and safely. 2.6 Delivered sliver is coiled uniformly into cans and cans are changed and identified correctly. 2.7 Stop motions are checked at intervals for proper functioning.
3. Perform lap preparation for combing	3.1 The lap former/unilap machine is prepared as per requirement. 3.2 The required number of slivers is creeled for <u>lap formation</u> . 3.3 <u>Lap parameters</u> are maintained during the operation.

Elements of Competency	Performance Criteria
	3.4 The lap is doffed, weighed and transferred to the comber without damage. 3.5 <u>Lap defects</u> are identified and reported to the authority.
4. Carry out combing operations	4.1 The comber is prepared and the lap is loaded and positioned correctly. 4.2 <u>Combing parameters</u> are set/verified according to the requirement. 4.3 The combing operation is carried out following standard operating procedures. 4.4 <u>Noil percentage</u> is checked and maintained within the standard. 4.5 <u>Comber faults</u> are identified and attended or reported. 4.6 Comber sliver is coiled into cans and identified properly. 4.7 Noil and waste are collected, segregated and recorded.
5. Check sliver quality and record process data	5.1 <u>Sliver quality parameters</u> are checked at specified intervals. 5.2 Hank/weight, unevenness and <u>sliver defects</u> are identified and recorded. 5.3 Corrective action is taken or reported as per workplace procedure. 5.4 Waste is collected, segregated, weighed and recorded. 5.5 Machine and work area are cleaned as per the cleaning schedule. 5.6 Data are recorded in the <u>prescribed format</u>.
6. Develop concept of Auto-leveler	6.1 The purpose and basic operating principle of an auto-leveler are explained. 6.2 Major components and functions of the auto-leveler are identified. 6.3 The role of auto-leveler in controlling sliver irregularity is explained. 6.4 Basic auto-leveler settings, monitoring requirements and safety precautions are identified
7. Handle machine and product faults, wastages, and cleaning	7.1 Machine and product faults are identified according to specified quality and operating requirements. 7.2 Appropriate corrective actions are taken to control machine and product faults. 7.3 Wastages are identified, segregated and handled according to workplace waste-management procedures 7.4 Machine cleaning is performed using appropriate tools and methods.

Range of Variables:

Variable	Range (May include but not limited to)
1. Drafting system parts	1.1 Top and bottom drafting rollers 1.2 Cots and aprons 1.3 Roller clearers and top clearer 1.4 Pressure/loading arms 1.5 Creel rollers and sliver guides 1.6 Trumpet, calender rollers and coiler 1.7 Suction/waste system
2. Roller settings and drafts	2.1 Total draft and tension draft 2.2 Front zone and back zone roller gauge 2.3 Top arm pressure setting 2.4 Trumpet size 2.5 Coiler and can plate setting
3. Doublings	3.1 6 or 8 doublings (typical practice) 3.2 Hank of feed sliver 3.3 Number of cans in the creel 3.4 Blend doubling for blended yarn 3.5 Can arrangement
4. Stop motions	4.1 Sliver break/feed stop motion 4.2 Can full stop motion 4.3 Roller lapping/lap-up stop motion 4.4 Pneumatic suction failure stop 4.5 Safety door interlock 4.6 Coiler choking stop
5. Auto-leveller	5.1 Open-loop and closed-loop levelling 5.2 Scanning/measuring rollers 5.3 Levelling action point 5.4 Correction length and intensity 5.5 Servo drive and control panel 5.6 Levelling monitoring and alarm
6. Creel, drafting zone, trumpet and coiler	6.1 Creel sliver guide and creel roller 6.2 Back zone draft 6.3 Main draft zone and front roller 6.4 Trumpet and calender rollers 6.5 Coiler tube and can plate 6.6 Suction pipe and waste box
7. Process parameters	7.1 Delivery speed (m/min) 7.2 Hank of delivered sliver 7.3 Draft and doubling 7.4 Production per shift and efficiency 7.5 Waste percentage 7.6 Relative humidity and temperature

Variable	Range (May include but not limited to)
8. Lap formation	8. 1 Super lap machine 8. 2 Number of slivers creeled 8. 3 Doubling and draft in lap preparation 8. 4 Lap winding tension
9. Lap parameters	9.1 Lap weight (g/m) 9.2 Lap width 9.3 Lap length per doff 9.4 Number of laps per doff 9.5 Hank of feed sliver
10. Lap defects	10.1 Lap licking and sticking 10.2 Splitting of lap layers 10.3 Uneven lap weight 10.4 Soft or hard lap 10.5 Damaged lap edges 10.6 Fluff attachment
11. Combing parameters	11.1 Feed length (mm) per nip 11.2 Type of feed (forward/backward) 11.3 Nips per minute 11.4 Detaching distance 11.5 Cylinder comb and top comb setting 11.6 Top comb penetration depth
12. Noil percentage	12. 1 Typical range of 12% to 22% depending on quality requirement 12. 2 Control by top comb penetration 12. 3 Short fibre
13. Comber faults	13. 1 Irregular or defective piecing 13. 2 High or low noil percentage 13. 3 Lap licking and splitting 13. 4 Damaged cylinder needles or top comb 13. 5 Uneven or fluffy web 13. 6 Sliver breakage at the table 13. 7 Fluff and fly accumulation 13. 8 Uneven sliver weight between heads
14. Sliver quality parameters	14. 1 Hank/count of sliver 14. 2 Weight per unit length 14. 3 Unevenness (U%/CV%) 14. 4 Short fibre content after combing 14. 5 Neps per gram 14. 6 Spectrogram fault/periodic variation 14. 7 Sliver appearance and coiling
15. Sliver defects	15. 1 Thick and thin places 15. 2 Uneven and irregular sliver

Variable	Range (May include but not limited to)
	15. 3 Single and double sliver 15. 4 Fibre hooks and poor parallelization 15. 5 Roller lapping marks 15. 6 Stretched or damaged sliver 15. 7 Contamination in sliver
16. Prescribed format	16. 1 Sliver hank check sheet 16. 2 Noil percentage record 16. 3 Shift production report 16. 4 Waste record 16. 5 Stoppage/downtime record 16. 6 Cleaning and maintenance checklist

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Objectives and functions of the drawing process – doubling, drafting, blending and parallelization. 1. 2 Construction and working principle of the draw frame and its drafting system. 1. 3 Roller settings, break draft, main draft and their effect on sliver quality 1. 4 Auto-levelling principle, types and its effect on count control. 1. 5 Objectives of combing and its effect on yarn quality. 1. 6 Lap preparation methods and the importance of lap quality. 1. 7 Combing parameters and their effect on noil percentage and yarn quality. 1. 8 Fibre hooks, short fibre removal and the sequence of drawing passages before and after combing. 1. 9 Sliver quality parameters, testing intervals and permissible tolerances. 1. 10 Cleaning schedules, housekeeping and record keeping.
2. Underpinning Skills	2.1 Checking drafting system parts, cots, aprons and clearers 2.2 Starting and stopping the draw frame and comber safely following SOPs 2.3 Piecing slivers and clearing roller lapping safely 2.4 Operating the lap former and maintaining lap parameters 2.5 Loading laps on the comber and setting combing parameters under supervision 2.6 Checking and controlling noil percentage 2.7 Identifying lap, comber and sliver defects and their probable causes 2.8 Measuring sliver hank/weight and checking evenness 2.9 Collecting, segregating, weighing and recording waste and noil

	2.10 Recording production, quality and maintenance data in prescribed formats
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Draw frame (breaker and finisher) with accessories 4.3 Lap former/unilap machine and comber machine 4.4 Card sliver, drawn sliver and lap material 4.5 Sliver cans, lap spools and material handling trolleys 4.6 Setting and feeler gauges 4.7 Electronic balance and sliver/lap weighing arrangement 4.8 Tachometer and hand lens 4.9 SOPs, machine setting sheets and recording formats 4.10 Learning manual and audio-visual clips

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 Prepared the draw frame for operation. 1. 2 Carried out breaker and finisher drawing operations. 1. 3 Performed lap preparation for combing. 1. 4 Carried out combing operations. 1. 5 Checked sliver quality and recorded process data. 1. 6 Developed concept of Auto-leveler 1. 7 Handled machine and product faults, wastages and cleaning
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM ROVING (SIMPLEX) OPERATIONS	Nominal Duration: 40 Hrs.	Unit Code: SICIP-TEX-PPYM-05-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform roving (simplex) operations. It specifically includes the tasks of preparing the speed frame for operation, carrying out the roving operation, performing creeling, piecing and doffing, and checking roving quality and maintaining records.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare the speed frame for operation	1. 1 Personal protective equipment (PPE) is selected and used in accordance with safety guidelines. 1. 2 Work order, lot details and the machine setting sheet are obtained and understood. 1. 3 <u>Machine parts</u> are checked for condition and cleanliness. 1. 4 <u>Settings and parameters</u> are set/verified as per the requirement. 1. 5 The creel is loaded with finisher drawing cans and slivers are threaded correctly. 1. 6 Empty <u>bobbins</u> are loaded and the bobbin rail is set at the correct starting position. 1. 7 Safety guards and stop motions are checked for proper functioning.
2. Carry out the roving operation	2.1 The machine is started and stopped following standard operating procedures. 2.2 Sliver is passed through the <u>drafting and twisting zone</u> correctly. 2.3 <u>Roving parameters</u> are maintained during the operation. 2.4 <u>Builder motion and package winding</u> are checked for correct bobbin build. 2.5 Roving breakages are attended promptly and correctly. 2.6 <u>Lapping</u> is removed safely from the drafting zone. 2.7 Stoppages are attended or reported according to the level of authority.
3. Perform creeling, piecing and doffing	3.1 Exhausted cans are replaced with full cans and sliver ends are joined correctly. 3.2 Roving <u>piecing</u> is done following the standard method. 3.3 <u>Doffing</u> is carried out safely at the correct time. 3.4 Full bobbins are handled, identified and transferred to the ring frame without damage.

Elements of Competency	Performance Criteria
	3.5 Empty bobbins are cleaned, checked for damage and arranged for reuse. 3.6 Bobbin trolleys and transport creels are arranged and used properly.
4. Check roving quality and maintain records	4.1 <u>Roving quality parameters</u> are checked at specified intervals. 4.2 Hank of roving and twist are measured and recorded. 4.3 <u>Roving defects</u> are identified and reported. 4.4 Waste is collected, segregated, weighed and recorded. 4.5 Machine and work area are cleaned as per the cleaning schedule. 4.6 Data are recorded in the <u>prescribed format</u>.

Range of Variables:

Variable	Range (May include but not limited to)
1. Machine parts	1. 1 Creel, sliver guide and creel rollers 1. 2 Drafting rollers with cots and aprons 1. 3 Spacer and condensers 1. 4 Top arm/pressure unit 1. 5 Flyer, presser arm and flyer top 1. 6 Spindle and bobbin holder 1. 7 Bobbin rail and builder motion 1. 8 Cone drum/servo drive 1. 9 Top clearer and suction unit 1. 10 Stop motion sensors
2. Settings and parameters	2.1 Break draft and main draft 2.2 Roller setting/gauge (front and back zone) 2.3 Spacer size 2.4 Top arm pressure 2.5 Twist per inch (TPI) 2.6 Spindle and flyer speed (rpm) 2.7 Bobbin and rail speed
3. Bobbins	3.1 Bobbin size and length 3.2 Empty and full bobbin 3.3 Damaged and bent bobbin identification 3.4 Colour coding of bobbins 3.5 Bobbin cleaning requirement
4. Drafting and twisting zone	4.1 Apron 4.2 Draft 4.3 Front roller and condenser 4.4 Flyer 4.5 Presser arm 4.6 Twist insertion zone between front roller and flyer

Variable	Range (May include but not limited to)
5. Roving parameters	5.1 Hank of roving (e.g., 0.60–1.20 Ne) 5.2 Twist multiplier (TM) and TPI 5.3 Spindle speed (rpm) 5.4 Package build, taper and lift 5.5 Production per shift and machine efficiency 5.6 Waste percentage
6. Builder motion and package winding	6.1 Layer-by-layer winding 6.2 Bobbin rail traverse and reversal 6.3 Taper formation at top and bottom 6.4 Ratchet/servo controlled build 6.5 Bobbin build faults (soft, hard, uneven)
7. Lapping	7.1 Front roller lapping 7.2 Apron and cot lapping 7.3 Choking of suction
8. Piecing	8.1 Overlap piecing method 8.2 Piecing length and strength 8.3 Avoidance of thick and thin piecing 8.4 Removal of damaged roving before piecing 8.5 Piecing at the correct position in the drafting zone
9. Doffing	9.1 Manual doffing 9.2 Semi-automatic and automatic doffing 9.3 Doffing interval and doff time 9.4 Safe handling and transport of full bobbins 9.5 Bobbin tray/trolley arrangement 9.6 Recording of doff details
10. Roving quality parameters	10.1 Hank of roving and weight per unit length 10.2 Twist per inch 10.3 Unevenness (U%/CV%) 10.4 Roving appearance and hairiness 10.5 Bobbin build, hardness and taper 10.6 Count variation between spindles
11. Roving defects	11.1 Uneven, thick or thin roving 11.2 High or low twist 11.3 Soft, hard or uneven bobbin build 11.4 Lapping and damaged roving 11.5 Missing, single or double roving 11.6 Roving with contamination
12. Prescribed format	12. 1 Hank of roving check sheet 12. 2 Twist test record 12. 3 Shift production report 12. 4 Waste record 12. 5 Doffing record

Variable	Range (May include but not limited to)
	12. 6 Stoppage/downtime record 12. 7 Cleaning and maintenance checklist

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Objectives and functions of the roving (simplex) process. 1. 2 Construction and working principle of the speed frame. 1. 3 Drafting system, apron drafting and the function of spacers and condensers. 1. 4 Roving parameters: hank, twist multiplier, spindle speed and stretch. 1. 5 Effect of excessive or insufficient twist on subsequent ring spinning. 1. 6 Roving quality parameters and permissible tolerances. 1. 7 Piecing methods and their effect on yarn faults. 1. 8 Doffing procedure, bobbin handling and transport practices. 1. 9 Types of waste at the speed frame and permissible limits. 1. 10 Cleaning schedules, housekeeping and record keeping requirements.
2. Underpinning Skills	2.1 Checking machine parts, cots, aprons and clearers before start-up 2.2 Setting and verifying draft, twist and speed parameters under supervision 2.3 Creeling finisher drawing cans and threading slivers correctly 2.4 Loading empty bobbins and setting the bobbin rail position 2.5 Starting and stopping the speed frame safely following SOPs 2.6 Performing roving piecing using the standard method 2.7 Clearing lap-ups and fluff safely 2.8 Carrying out doffing and handling full bobbins without damage 2.9 Measuring hank of roving and twist per inch 2.10 Identifying roving defects and their probable causes 2.11 Recording production, quality and maintenance data
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace

	3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Speed frame (simplex) with accessories 4.3 Finisher drawing sliver and cans 4.4 Empty and full roving bobbins, bobbin trolleys 4.5 Setting and feeler gauges, spacers 4.6 Electronic balance and wrap reel/roving weighing arrangement 4.7 Twist tester 4.8 Tachometer and hand lens 4.9 SOPs, machine setting sheets and recording formats 4.10 Learning manual and audio-visual clips

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 Prepared the speed frame for operation. 1. 2 Carried out the roving operation. 1. 3 Performed creeling, piecing and doffing. 1. 4 Checked roving quality and maintained records.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM RING SPINNING OPERATIONS	Nominal Duration: 60 Hrs.	Unit Code: SICIP-TEX-PPYM-06-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform ring spinning operations. It specifically includes the tasks of preparing the ring frame for operation, carrying out the ring spinning operation, performing piecing, creeling and doffing, monitoring yarn quality and controlling end breakages, and handling waste, cleaning and shift records.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare the ring frame for operation	1. 1 Personal protective equipment (PPE) is selected and used in accordance with safety guidelines. 1. 2 Work order, lot details and the <u>setting parameters</u> are obtained and understood. 1. 3 <u>Ring frame parts</u> are checked for condition, cleanliness and alignment. 1. 4 <u>Spinning settings</u> are set/verified as per the requirement. 1. 5 <u>Travellers, rings and cots</u> are checked and replaced as required. 1. 6 Roving bobbins are creeled correctly and empty tubes are loaded on the spindles. 1. 7 Safety guards, pneumafil suction and stop motions are checked for proper functioning.
2. Carry out the ring spinning operation	2.1 The machine is started and stopped following standard operating procedures. 2.2 Roving is passed through the <u>drafting zone, lappet, traveller and ring</u> correctly. 2.3 <u>Process parameters</u> are maintained during the operation. 2.4 Yarn is wound onto the cop with the correct <u>package build</u> . 2.5 Running of all spindles is monitored regularly during the shift. 2.6 Idle spindles and <u>lapping</u> are attended promptly and safely. 2.7 Pneumafil waste is checked and controlled within the standard.
3. Perform piecing, creeling and doffing	3.1 <u>Piecing</u> of broken ends is carried out correctly and promptly. 3.2 Creeling of exhausted roving bobbins is done following the standard method. 3.3 <u>Doffing</u> is carried out safely at the scheduled time.

Elements of Competency	Performance Criteria
	3.4 Cops are collected, handled and transferred without damage or contamination. 3.5 Empty bobbins and tubes are cleared, cleaned and stored properly. 3.6 Doff details are recorded as per workplace procedure.
4. Monitor yarn quality and control end breakages	4.1 <u>Yarn quality parameters</u> are checked at specified intervals. 4.2 <u>End breakage rate</u> is measured and recorded as per standard practice. 4.3 <u>Yarn defects</u> and their probable causes are identified. 4.4 Count and twist are verified against the specification. 4.5 Corrective action is taken or reported to the authority. 4.6 Data are recorded in the <u>prescribed format</u> .
5. Handle waste, cleaning and shift records	5.1 Waste is collected, segregated, weighed and recorded following workplace procedure. 5.2 <u>Cleaning</u> of the machine, creel, drafting zone and under-machine area is carried out as per schedule. 5.3 <u>Workplace atmospheric conditions</u> are checked and abnormalities are reported. 5.4 Production, efficiency and downtime records are completed accurately. 5.5 The machine is handed over to the next shift with proper information.

Range of Variables:

Variable	Range (May include but not limited to)
1. Setting parameters	1.1 Lot number and yarn count 1.2 Twist per inch and twist multiplier 1.3 Spindle speed (rpm) 1.4 Total draft and break draft 1.5 Spacer size and top arm pressure 1.6 Traveller number and type 1.7 Ring cup size 1.8 Cop weight, traverse length and build 1.9 Production and efficiency target
2. Ring frame parts	2.1 Creel and roving guide/bobbin holder 2.2 Drafting rollers with cots and aprons 2.3 Spacer and condenser 2.4 Top arm/pressure unit 2.5 Lappet and balloon control ring 2.6 Ring, ring rail and traveller 2.7 Spindle and separator 2.8 Builder/lifting motion

Variable	Range (May include but not limited to)
	2.9 Pneumafil suction and waste box 2.10 Top clearer and roller clearers 2.11 Drive motor, inverter and gearing
3. Spinning settings	3.1 Break draft and main draft 3.2 Roller gauge/setting 3.3 Spacer size 3.4 Top arm pressure 3.5 Twist per inch 3.6 Spindle speed 3.7 Ring rail lift, traverse length 3.8 Balloon control ring position 3.9 Lappet height
4. Travellers, rings and cots	4.1 Traveller number, type and profile 4.2 Traveller change schedule and running-in period 4.3 Ring type, condition and ring run-in 4.4 Cot hardness 4.5 Cot mounting 4.6 Apron condition and thickness
5. Drafting zone, lappet, traveller and ring	5.1 Back zone and main zone Draft 5.2 Apron 5.3 Lappet 5.4 Balloon control ring 5.5 Traveller running on the ring 5.6 Spindle and cop formation
6. Process parameters	6.1 Spindle speed (rpm) 6.2 Twist per inch and twist direction (S/Z) 6.3 Draft and yarn count 6.4 Delivery rate and production per shift 6.5 Cop weight and build 6.6 Machine efficiency percentage 6.7 End breakage rate 6.8 Pneumafil waste percentage 6.9 Relative humidity and temperature
7. Package build	7.1 Cop build 7.2 Layer winding and traverse 7.3 Soft, hard or uneven cop 7.4 Cop weight
8. Lapping	8.1 Front roller lapping 8.2 Back roller and apron lapping 8.3 Cot lapping due to damaged or dirty cots 8.4 Roving lap in the creel
9. Piecing	9.1 Hand piecing method

Variable	Range (May include but not limited to)
	9.2 Piecing strength and appearance 9.3 Avoidance of thick piecing and double ends 9.4 Correct piecing time to reduce spindle idleness
10. Doffing	10.1 Manual doffing 10.2 Automatic doffing 10.3 Doffing interval and doff cycle time 10.4 Cop handling using trays and trolleys 10.5 Identification and segregation of doffed cops by lot
11. Yarn quality parameters	11.1 Count (Ne/tex) and count variation (CV%) 11.2 Twist per inch 11.3 Single yarn strength, CSP and RKM 11.4 Elongation percentage 11.5 Unevenness (U%/CV%) 11.6 Imperfections (thin places, thick places, neps per km) 11.7 Hairiness index
12. End breakage rate	12. 1 Breaks per 100 spindle hours 12. 2 Method of counting breakages 12. 3 Material related causes 12. 4 Machine and setting related causes 12. 5 Atmospheric (humidity/temperature) related causes 12. 6 Operator/handling related causes
13. Yarn defects	13. 1 Slubs and thick places 13. 2 Thin places and weak spots 13. 3 Neps and hairiness 13. 4 Count variation between spindles 13. 5 Double and single ends 13. 6 Oil, dirt and grease stains 13. 7 Foreign fibre and contamination 13. 8 Bad piecing and cop stitching
14. Cleaning	1.4.1 Creel and roving guide cleaning 1.4.2 Drafting zone, cot and apron cleaning 1.4.3 Ring, traveller and ring rail cleaning 1.4.4 Spindle and separator cleaning 1.4.5 Pneumafil and suction duct cleaning 1.4.6 Under-machine and floor cleaning 1.4.7 Filter cleaning and vacuum stripping 1.4.8 Overhead traveller cleaner (OHTC) operation
15. Workplace atmospheric conditions	15. 1 Relative humidity (generally 50–60% in the ring frame department) 15. 2 Temperature (generally 25–32°C) 15. 3 Air circulation and ventilation 15. 4 Fly and dust level 15. 5 Lighting level

Variable	Range (May include but not limited to)
	15. 6 Noise level
16. Prescribed format	16. 1 Shift production and efficiency report 16. 2 Count and twist check sheet 16. 3 End breakage record 16. 4 Waste record (pneumafil, sweep, hard waste) 16. 5 Doffing record 16. 6 Downtime/stoppage record 16. 7 Cleaning and maintenance checklist 16. 8 Shift handover log book

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Objectives and functions of the ring spinning process. 1.2 Construction and working principle of the ring frame. 1.3 Twist insertion by ring and traveller and the concept of twist multiplier. 1.4 Function of lappet, balloon control ring, separator and builder motion. 1.5 Selection of traveller number in relation to yarn count and spindle speed. 1.6 Yarn quality parameters, testing methods and permissible tolerances. 1.7 Piecing methods and their effect on yarn faults and productivity. 1.8 Doffing procedure, cop handling and lot segregation. 1.9 Importance of relative humidity and temperature control in spinning. 1.10 Machine efficiency, production calculation and downtime analysis. 1.11 Cleaning schedules, housekeeping and shift handover practices.
2. Underpinning Skills	2.1 Checking ring frame parts, cots, aprons, rings and travellers before start-up 2.2 Verifying draft, twist, spindle speed and spacer settings under supervision 2.3 Changing travellers and cots as per schedule 2.4 Creeling roving bobbins and threading roving through the drafting zone 2.5 Starting and stopping the ring frame safely following SOPs 2.6 Performing piecing correctly using the suction gun and by hand 2.7 Carrying out doffing and handling cops without damage 2.8 Measuring yarn count and twist and checking cop weight

	2.9 Identifying yarn defects and their probable causes 2.10 Collecting, segregating, weighing and recording waste 2.11 Checking humidity and temperature and reporting abnormalities
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Ring frame with accessories 4.3 Traveller sets, traveller gauge and cot buffing machine 4.4 Setting and feeler gauges, spacers 4.5 Suction/piecing gun 4.6 Wrap reel, wrap block and electronic balance 4.7 Twist tester and lea strength tester 4.8 Tachometer, hygrometer and thermometer 4.9 Cleaning tools, vacuum cleaner and OHTC 4.10 SOPs, setting sheets and recording formats 4.11 Learning manual and audio-visual clips

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Prepared the ring frame for operation. 1.2 Carried out the ring spinning operation. 1.3 Performed piecing, creeling and doffing. 1.4 Monitored yarn quality and controlled end breakages. 1.5 Handled waste, cleaning and shift records.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM WINDING AND YARN PACKING OPERATIONS	Nominal Duration: 40 Hrs.	Unit Code: SICIP-TEX-PPYM-07-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform winding and yarn packing operations. It specifically includes the tasks of preparing the winding machine for operation, carrying out winding operations, performing splicing/knotting and yarn clearing, and carrying out conditioning, packing and dispatch preparation.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare the winding machine for operation	1.1 Work order and lot details are obtained and understood. 1.2 <u>Winding machine parts</u> are checked for condition and cleanliness. 1.3 <u>Winding settings</u> are set/verified as per the requirement. 1.4 <u>Cones and paper tubes</u> and ring cops are arranged as per requirement. 1.5 The <u>yarn clearer and splicer</u> are checked for proper functioning. 1.6 Safety guards and stop motions are checked before start-up.
2. Carry out winding operations	2.1 The machine is started and stopped following standard operating procedures. 2.2 Cops are loaded/creeled correctly and yarn is threaded through the <u>yarn path</u> . 2.3 <u>Winding parameters</u> are maintained during the operation. 2.4 <u>Package build and density</u> are checked at intervals. 2.5 Yarn breakages and stoppages are attended promptly. 2.6 Hard waste is collected and controlled within the standard. 2.7 Drum, tension device and yarn path are kept clean and free from fluff.
3. Perform splicing and yarn clearing	3.1 <u>Splicing</u> is performed following the standard method. 3.2 Splice/knot strength and appearance are checked against the requirement. 3.3 <u>Yarn clearer settings</u> are verified for removal of objectionable faults. 3.4 <u>Faults</u> are identified and recorded. 3.5 Clearer cuts and machine efficiency are monitored and reported.

Elements of Competency	Performance Criteria
	3.6 Abnormal cutting or malfunction of the clearer/splicer is reported to the authority.
4. Carry out conditioning, packing and dispatch preparation	4.1 Cones are inspected, weighed and identified with <u>lot and count details</u>. 4.2 <u>Yarn conditioning</u> requirements are followed where applicable. 4.3 Cones are packed in cartons/bags following <u>packing specifications</u>. 4.4 <u>Labelling and documentation</u> are completed accurately. 4.5 Packed materials are stacked and stored safely in the designated area. 4.6 Work area is cleaned and records are completed as per workplace procedure.

Range of Variables:

Variable	Range (May include but not limited to)
1. Winding machine parts	1.1 Creel/cop holder and cop stand 1.2 Yarn guide 1.3 Tension device (disc/gate type) 1.4 Yarn clearer (electronic/mechanical) 1.5 Splicer unit and air supply 1.6 Waxing device 1.7 Grooved drum and cradle 1.8 Cone holder and package sensor 1.9 Suction/waste system 1.10 Auto doffing arm (in autoconer)
2. Winding settings	2.1 Winding speed (m/min) 2.2 Yarn tension setting 2.3 Cradle pressure and package hardness 2.4 Drum groove type and traverse 2.5 Waxing pressure and wax type 2.6 Package density and taper 2.7 Package length/weight setting
3. Cones and paper tubes	3.1 Taper angle (3°30', 4°20', 5°57', 9°15') 3.2 Paper cone and dyeing cone 3.3 Cone length and weight 3.4 Colour coding of cones by count/lot 3.5 Damaged cone identification
4. Yarn clearer and splicer	4.1 Clearer sensor and control panel 4.2 Splicer chamber and opening tubes 4.3 Splicer air pressure

Variable	Range (May include but not limited to)
	4.4 Opening time and twisting time 4.5 Thread cutter and suction nozzle
5. Yarn path	5.1 Cop → balloon breaker → tension device 5.2 Tension device → yarn clearer 5.3 Clearer → splicer → waxing device 5.4 Waxing device → grooved drum → cone 5.5 Suction nozzle for upper and lower yarn
6. Winding parameters	6.1 Winding speed and tension 6.2 Package weight and density 6.3 Splice quality and strength 6.4 Machine efficiency percentage 6.5 Hard waste percentage 6.6 Number of packages per shift
7. Package build and density	7.1 Cross wound and precision wound package 7.2 Package hardness and density 7.3 Nose and base formation
8. Splicing	8.1 Pneumatic (air) splicing 8.2 Splice strength (generally 80–90% of yarn strength) 8.3 Splice testing method
9. Yarn clearer settings	9.1 Thin place channel 9.2 Thick place (short and long) channel 9.3 Nep channel 9.4 Long thin fault channel 9.5 Count channel/count deviation 9.6 Foreign fibre channel
10. Faults	10.1 Slubs and thick places 10.2 Thin places and weak spots 10.3 Neps 10.4 Spinner's doubles 10.5 Foreign fibre and contamination 10.6 Count variation faults 10.7 Bad piecing from the ring frame
11. Lot and count details	11.1 Lot number and mixing number 11.2 Yarn count and blend 11.3 Machine and shift number 11.4 Date of production 11.5 Gross weight and net weight 11.6 Shade/batch identification where applicable
12. Yarn conditioning	12.1 Steam/vacuum conditioning chamber 12.2 Conditioning time and temperature 12.3 Target moisture regain 12.4 Setting of twist by conditioning

Variable	Range (May include but not limited to)
	12. 5 Handling of cones before and after conditioning
13. Packing specifications	13. 1 Number of cones per carton/bag 13. 2 Net weight per bag/carton (e.g., 45.36 kg) 13. 3 Poly bag and carton size and quality 13. 4 Protection against moisture and contamination
14. Labelling and documentation	14. 1 Lot and count label on the cone 14. 2 Carton/bag marking and sticker 14. 3 Barcode/QR code where applicable 14. 4 Packing list and delivery challan 14. 5 Store and dispatch register 14. 6 Weight and inspection record

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Construction and working principle of the drum winder and autoconer. 1.2 Yarn path, tension devices and balloon control in winding. 1.3 Package building principle, cross winding and package density. 1.4 Causes and prevention of ribboning, stitching and sloughing off. 1.5 Principle and working of electronic yarn clearers and their channels. 1.6 Splicing principle, splice quality requirements and knot types. 1.7 Yarn conditioning purpose, method and effect on moisture regain and twist setting. 1.8 Packing specifications, labelling, weighing and storage requirements. 1.9 Types of waste in winding (hard waste) and permissible limits. 1.10 Machine efficiency, clearer cuts and productivity monitoring. 1.11 Documentation for production, packing and dispatch.
2. Underpinning Skills	2. 1 Checking winding machine parts, clearer and splicer before start-up 2. 2 Loading cops and threading yarn through the correct yarn path 2. 3 Starting and stopping the winding machine safely following SOPs 2. 4 Monitoring package build and density during winding 2. 5 Performing splicing and knotting using the standard method

	2. 6 Checking splice strength and appearance 2. 7 Verifying and interpreting yarn clearer settings and clearer cuts 2. 8 Inspecting, weighing and identifying finished cones 2. 9 Carrying out yarn conditioning where applicable 2. 10 Packing, labelling, stacking and storing cones as per specification 2. 11 Completing packing, weighing and dispatch documentation 2. 12 Collecting, segregating and recording hard waste
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4. 1 Workplace (simulated or actual) 4. 2 Winding machine (drum winder/autoconer) with accessories 4. 3 Ring cops, paper cones and tubes 4. 4 Splicer accessories and air supply arrangement 4. 5 Yarn clearer with control panel 4. 6 Electronic balance and weighing scale 4. 7 Yarn strength tester for splice testing 4. 8 Yarn conditioning arrangement (where available) 4. 9 Cartons, poly bags, strapping and labelling materials 4. 10 Learning manual and audio-visual clips

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Prepared the winding machine for operation. 1.2 Carried out winding operations. 1.3 Performed splicing/knotting and yarn clearing. 1.4 Carried out conditioning, packing and dispatch preparation.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)

3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.
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Unit of Competency: PERFORM IN-PROCESS QUALITY CHECKS AND ROUTINE MACHINE CARE	Nominal Duration: 20 Hrs.	Unit Code: SICIP-TEX-PPYM-08-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform in-process quality checks and routine machine care. It specifically includes the tasks of performing in-process quality checks and basic tests, recording and reporting production and quality data, carrying out cleaning and routine preventive maintenance, and maintaining housekeeping and workplace conditions.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Perform in-process quality checks and basic tests	1.1 The <u>sampling procedure</u> is followed at specified intervals. 1.2 <u>Basic quality tests</u> are performed using appropriate instruments. 1.3 <u>Quality parameters</u> are checked against the standard. 1.4 Out-of-standard results are identified and reported promptly. 1.5 Test samples are labelled, handled and preserved as per procedure. 1.6 Testing instruments are handled carefully and kept clean.
2. Record and report production and quality data	2. 1 Production, waste and quality data are recorded in the <u>prescribed formats</u> . 2. 2 Simple calculations are performed correctly where required. 2. 3 <u>Reports</u> are prepared and submitted to the authority within the required time. 2. 4 Abnormalities and breakdowns are reported through the correct <u>reporting channel</u> . 2. 5 Records are kept clean, legible, complete and traceable.
3. Carry out cleaning and routine preventive maintenance	3.1 <u>Routine maintenance activities</u> are carried out as per schedule and level of authority. 3.2 <u>Lubrication and cleaning points</u> are attended following the maintenance chart. 3.3 Worn parts and abnormal conditions are identified and reported. 3.4 <u>Safety procedures</u> are followed during maintenance work. 3.5 Tools are selected, used correctly and returned to the designated place. 3.6 Maintenance activities are recorded in the checklist.

Elements of Competency	Performance Criteria
4. Maintain housekeeping and workplace conditions	4. 1 <u>5S and housekeeping practices</u> are applied in the work area. 4. 2 <u>Workplace conditions</u> are checked and abnormalities are reported. 4. 3 Materials, tools and waste are stored in designated places. 4. 4 Walkways, machine aisles and emergency exits are kept clear. 4. 5 Waste is segregated and disposed of as per the environmental procedure. 4. 6 The work area is handed over in a clean and safe condition.

Range of Variables:

Variable	Range (May include but not limited to)
1. Sampling procedure	1.1 Sampling frequency (per shift, per machine, per lot) 1.2 Number and size of samples 1.3 Sampling location and method 1.4 Sample identification and labelling 1.5 Conditioning of samples before testing 1.6 Standard atmospheric condition for testing (20±2°C, 65±2% RH)
2. Basic quality tests	2. 1 Lap, sliver and roving weight per unit length 2. 2 Yarn count using wrap reel and electronic balance 2. 3 Twist test (TPI) using twist tester 2. 4 Lea strength/CSP and single yarn strength test 2. 5 Unevenness (U%/CV%) using evenness tester 2. 6 Imperfection and hairiness measurement 2. 7 Trash and noil percentage 2. 8 Moisture regain measurement 2. 9 Nep counting and appearance grading
3. Quality parameters	3.1 Hank/count and count variation 3.2 Unevenness (U%/CV%) 3.3 Imperfections per kilometre 3.4 Strength (CSP/RKM) and elongation 3.5 Twist per inch 3.6 Waste and noil percentage 3.7 End breakage rate 3.8 Package weight and build 3.9 Moisture regain
4. Prescribed formats	4. 1 Shift and daily production report 4. 2 Hank/count check sheet 4. 3 Waste report

Variable	Range (May include but not limited to)
	4. 4 End breakage record 4. 5 Downtime/stoppage record 4. 6 Quality test report 4. 7 Maintenance checklist 4. 8 5S/housekeeping audit sheet
5. Reports	5.1 Shift report and daily production report 5.2 Breakdown and downtime report 5.3 Machine handover report
6. Reporting channel	6.1 Operator → line leader/supervisor 6.2 Supervisor → shift in-charge 6.3 Shift in-charge → production/quality manager 6.4 Verbal and written reporting 6.5 Log book and digital/computerized reporting 6.6 Emergency reporting procedure
7. Routine maintenance activities	7.1 Cleaning and stripping of machines 7.2 Lubrication and greasing as per chart 7.3 Belt, pulley and chain checking and tensioning 7.4 Cot and traveller changing schedule 7.5 Filter and suction duct cleaning 7.6 Pneumatic line and air leakage checking 7.7 Checking of stop motions and safety devices 7.8 Setting and alignment checks by authorized personnel
8. Lubrication and cleaning points	8.1 Gear box, bearings and bushes 8.2 Spindle bolster and spindle oil level 8.3 Drafting rollers and top arm units 8.4 Chains, belts and pulleys 8.5 Suction ducts, filters and waste boxes 8.6 Grooved drum, tension devices and guides 8.7 Creel rollers and sliver guides
9. Safety procedures	9.1 Isolation of electrical power supply 9.2 Use of appropriate PPE 9.3 Permit-to-work system where applicable 9.4 Machine guarding and restoration after work 9.5 Confirmation and clearance before restart
10. 5S and housekeeping practices	10.1 Sort (Seiri) 10.2 Set in order (Seiton) 10.3 Shine (Seiso) 10.4 Standardize (Seiketsu) 10.5 Sustain (Shitsuke) 10.6 Floor marking and colour coding 10.7 Tool board and designated storage 10.8 Waste bin segregation

Variable	Range (May include but not limited to)
11. Workplace conditions	11.1 Relative humidity in each department 11.2 Temperature and air circulation 11.3 Lighting level at the workstation 11.4 Dust and fly level 11.5 Noise level 11.6 Ventilation and exhaust condition 11.7 Cleanliness of floor, passageways and emergency exits

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Importance of in-process quality control in yarn manufacturing. 1.2 Basic testing instruments, their use, care and calibration requirements. 1.3 Methods of testing count, twist, strength, evenness and imperfections. 1.4 Quality parameters, standards and permissible tolerances at each process stage. 1.5 Production, waste and efficiency calculation methods. 1.6 Concept and importance of preventive maintenance. 1.7 Correct selection, use and care of hand tools. 1.8 Concept and application of 5S and housekeeping in a spinning mill. 1.9 Effect of humidity, temperature, dust and lighting on quality, productivity and health. 1.10Waste segregation and environmentally sound disposal practices.
2. Underpinning Skills	2.1 Performing basic quality tests using appropriate instruments 2.2 Measuring count, twist, strength, weight per unit length and moisture 2.3 Comparing test results against standards and identifying deviations 2.4 Recording production, waste and quality data in prescribed formats 2.5 Performing simple production, waste and efficiency calculations 2.6 Preparing & submitting shift & quality reports, and carrying out routine cleaning & lubrication as per schedule 2.7 Applying lockout–tagout and safety procedures during maintenance 2.8 Selecting, using and storing hand tools correctly 2.9 Applying 5S and housekeeping practices in the work area 2.10Segregating and disposing of waste as per environmental procedure

3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Testing laboratory with wrap reel, electronic balance, twist tester and strength tester 4.3 Moisture meter, hygrometer and thermometer 4.4 Hand lens and nep counting board 4.5 Hand tool set, oil can and grease gun 4.6 Lubrication chart, maintenance checklist and cleaning schedule 4.7 Colour coded waste bins and storage arrangement 4.8 Sample lap, sliver, roving and yarn 4.9 SOPs, standards and recording formats 4.10 Learning manual and audio-visual clips

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Performed in-process quality checks and basic tests. 1.2 Recorded and reported production and quality data. 1.3 Carried out cleaning and routine preventive maintenance. 1.4 Maintained housekeeping and workplace conditions.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

WORKSHOP / LAB FACILITY STANDARD

Course Name:	Production Process in Yarn Manufacturing
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/Lab/Classroom cum Workshop):

1. Classroom – 500 sft (46.45 sqm)
2. Workshop/Lab – 1500 sft (140 sqm), OR
3. Classroom cum workshop – 1800 sft (167 sqm)

Major Training Equipment and Training Facilities:

S.N.	Major Equipment and Training Facilities	Required Quantity
1.	Computer/Laptop	1
2.	Multimedia projector & screen	1
3.	Printer	1
4.	Scanner	1
5.	Internet connection with Wi-Fi facility	1
6.	Whiteboard with accessories	1
7.	Bale opener/plucker with mixing arrangement	1
8.	Blow room line (opener, cleaner, mixer, condenser/chute feed)	1 set
9.	Carding machine	1
10.	Draw frame (breaker and finisher)	2
11.	Lap former/Unilap machine	1
12.	Comber machine	1
13.	Speed frame (Simplex)	1
14.	Ring frame	1
15.	Winding machine (drum winder/autoconer)	1
16.	Humidification unit/yarn conditioning arrangement	1
17.	Wrap reel with wrap block	2
18.	Electronic balance (0.001 g and 0.01 g accuracy)	1
19.	Lea strength tester/single yarn strength tester	1
20.	Twist tester	1
21.	Evenness tester (U%/CV%)	1
22.	Trash analyser	1

S.N.	Major Equipment and Training Facilities	Required Quantity
23.	Moisture meter	1
24.	Hygrometer and thermometer set	2
25.	Tachometer	1
26.	Feeler gauge and setting gauge set	5
27.	Hand lens/magnifier with nep counting board	5
28.	Hand tool set (spanner, screwdriver, plier, nipper, hook)	5
29.	Cot buffing machine	1
30.	Traveller gauge and traveller set (assorted)	2 sets
31.	Sliver cans, roving bobbins, ring tubes/cops, paper cones	As required
32.	Material handling trolley for cans, bobbins and cones	4
33.	Industrial vacuum cleaner/air blowing gun	2
34.	Waste segregation bins (colour coded)	4
35.	Tool board and storage cabinet	2
36.	Fire extinguisher	2
37.	First aid box	1
38.	PPE set (dust mask, ear plug, gloves, apron, cap, safety shoes)	25 sets

The following conditions must be fulfilled –

1. The institute shall not use the same facilities for any other projects/organizations offering a similar course.
2. The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facilities required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirements of the Competency Standard (CS) provided by SICIP.

For the operation of the training course on **Production Process in Yarn Manufacturing**, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

Where a training institute does not possess the full-scale production machinery, the institute must establish a formal linkage with a spinning mill so that the practical sessions of the occupation-specific units can be conducted in an actual workplace under supervision. The institute will also provide all other hand tools and power tools as per the CS for 25 trainees. The institute will also arrange adequate seating arrangements and classroom setup for the 25 trainees.